

Intelligent Phishing URL Classification System

A machine learning pipeline that classifies URLs as **legitimate** or **phishing** in real-time using structural, abnormal, and identity-based heuristics.

 Project Overview

Field	Detail
Project Name	CyberSentinel
Type	Binary Classification (Phishing Detection)
Language	Python 3.12
Environment	Google Colab / Local
Models	Random Forest · SVM (RBF kernel)
Dataset	Built-in demo · PhishTank · UCI Phishing Dataset

 Test Run Results

Pipeline executed successfully on **2025-09-25** using the built-in 10-URL demo dataset. All 7 pipeline stages completed with **zero critical errors**.

 Environment & Fixes Applied

Two issues were caught and fixed during testing:

Issue	Fix Applied
<code>tlldextract</code> attempted to fetch Public Suffix List from the web (403 Forbidden in sandbox)	Switched to offline snapshot mode: <code>TLDExtract(suffix_list_urls=[])</code>
Seaborn <code>boxplot</code> deprecated <code>palette</code> dict without <code>hue</code> in v0.14	Added <code>hue='label'</code> and <code>legend=False</code> per new API

Both fixes are already applied in `cybersentinel_pipeline.py`. No action needed.

Dataset Summary

```
Records loaded   : 10
Phishing URLs    : 6  (60%)
Legitimate URLs  : 4  (40%)
Features extracted : 28
Feature matrix shape : (10, 28)
```

Feature statistics (sample):

Feature	Mean	Std Dev
url_length	38.6	9.36
hostname_length	18.0	11.67
path_length	12.4	5.32
num_dots	2.4	1.17
num_slashes	3.8	0.79
num_hyphens	0.3	0.67

Random Forest — Results

```
              precision    recall  f1-score   support

 Legitimate      1.00      1.00      1.00         1
  Phishing      1.00      1.00      1.00         1

   accuracy              1.00         2
  macro avg      1.00      1.00      1.00         2
weighted avg      1.00      1.00      1.00         2

ROC-AUC   : 1.0000
CV F1     : 1.0000 ± 0.0000  (5-fold cross-validation)
```

◆ SVM (RBF Kernel) — Results

	precision	recall	f1-score	support
Legitimate	1.00	1.00	1.00	1
Phishing	1.00	1.00	1.00	1
accuracy			1.00	2
macro avg	1.00	1.00	1.00	2
weighted avg	1.00	1.00	1.00	2
ROC-AUC : 1.0000				
CV F1 : 0.8000 ± 0.4000 (5-fold cross-validation)				

Note: 100% accuracy on the test set is expected with a 10-sample demo dataset — it is not representative of real-world performance. When trained on PhishTank (~25k URLs), expect accuracy in the **95-97%** range. The SVM CV variance (± 0.4) reflects the small sample size.

🏆 Top 5 Feature Importances (Random Forest)

Rank	Feature	Importance Score
1	num_digits_in_url	0.1891
2	brand_keyword_in_path	0.1343
3	uses_https	0.0933
4	num_dots	0.0871
5	subdomain_depth	0.0761

● ● Live Prediction Demo — All 6 URLs

#	Verdict	Confidence	URL
1	● LEGITIMATE	26.2%	https://www.google.com/search?q=machine+learning
2	● PHISHING	84.6%	http://192.168.10.5/paypal/verify/account
3	● PHISHING	75.2%	http://secure.g00gle.com.phish.ru@evil.biz/login
4	● LEGITIMATE	31.4%	https://github.com/openai/gpt-4
5	● PHISHING	59.4%	http://paypa1-secure-login.xyz/update/credentials
6	● LEGITIMATE	48.7%	https://www.amazon.co.uk/dp/B0BTDVMZ9H

Prediction accuracy on demo set: 6/6 correct (100%)

Red flags triggered per phishing URL:

- <http://192.168.10.5/paypal/verify/account>
 - ■ Raw IP address as host
 - ■ Typosquatting characters found
 - ■ Brand keyword in path
 - ■ No HTTPS
- <http://secure.g00gle.com.phish.ru@evil.biz/login>
 - ■ @ symbol detected
 - ■ Brand keyword in path
 - ■ No HTTPS
- <http://paypa1-secure-login.xyz/update/credentials>
 - ■ Typosquatting characters found
 - ■ Suspicious TLD ([.xyz](#))
 - ■ Brand keyword in path
 - ■ No HTTPS

Output Plots Generated

All four plots were saved successfully during the test run:

File	Description
<code>cybersentinel_eda.png</code>	Class distribution · URL length boxplot · Feature correlation heatmap
<code>cybersentinel_confusion.png</code>	Confusion matrices for RF and SVM side by side
<code>cybersentinel_roc.png</code>	Overlaid ROC curves for both models
<code>cybersentinel_features.png</code>	Top 15 feature importances (Random Forest)

Project Structure

```
CyberSentinel/  
├─ cybersentinel_pipeline.py  # Full ML pipeline (feature extraction → prediction)  
├─ README.md                 # This file  
├─ cybersentinel_eda.png     # EDA plots  
├─ cybersentinel_confusion.png # Confusion matrices  
├─ cybersentinel_roc.png     # ROC curves  
└─ cybersentinel_features.png # Feature importances
```

Quick Start (Google Colab)

Step 1 — Install dependencies

```
python  
  
!pip install tldextract scikit-learn pandas seaborn matplotlib
```

Step 2 — Run on demo data

```
python  
  
exec(open('cybersentinel_pipeline.py').read())
```

Step 3 — Run on your own dataset (PhishTank CSV)

```
python  
  
df = load_dataset('phishtank_urls.csv')  # expects 'url' and 'label' columns  
features_df = build_feature_matrix(df['url'].tolist())
```

Step 4 — Predict any URL

```
python
```

```
result = predict_url('http://paypa1.com-secure.xyz/login', best_model)
print(result)
# {'verdict': 'PHISHING', 'confidence': 91.3, 'flags': [...]}
```

Heuristics Reference

Category	Feature	Red Flag Condition
Structural	num_at_symbols	> 0
Structural	num_dots	> 4
Structural	subdomain_depth	≥ 3
Structural	url_length	> 75 chars
Abnormal	is_ip_address	Hostname is raw IPv4/IPv6
Abnormal	has_port	Non-standard port in URL
Identity	typo_char_count	Any digit substituting a letter
Identity	has_brand_in_subdomain	Brand name buried in subdomain
Identity	suspicious_tld	.xyz, .tk, .ml, .click, etc.
Obfuscation	has_hex_encoding	%-encoded characters in URL
Obfuscation	url_shortener	Known shortener domain (bit.ly, etc.)

Dataset Sources

Source	URL	Notes
PhishTank	https://phishtank.org/developer_info.php	Free verified phishing URLs, updated daily
UCI Phishing Dataset	https://archive.ics.uci.edu/dataset/327	11,000+ URLs, pre-labeled
OpenPhish	https://openphish.com	Real-time phishing feed

⚠ Limitations

- Demo results (100% accuracy) are based on 10 URLs — **not production metrics**.
 - `tldextract` runs in offline mode in sandboxed environments; suffix detection may miss very new TLDs.
 - SVM cross-validation variance is high on small datasets — will stabilise with 10k+ samples.
 - Model does not yet use **domain age** (WHOIS) or **page content** features — planned for v2.
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📖 Roadmap

- ☐ Integrate WHOIS lookup for domain age feature
 - ☐ Add XGBoost / LightGBM as third model
 - ☐ Build Flask/FastAPI REST endpoint for real-time scoring
 - ☐ Deploy as Chrome Extension using the trained model
 - ☐ Add SHAP explainability for per-prediction feature attribution
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Generated by CyberSentinel test run · Python 3.12 · scikit-learn 1.x · September 2025